

Computer Science & IT

Discrete Mathematics

Mathematical Logic

Lecture No. 05

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Recap of Previous Lecture



- ✓ **Topic** Predicate logic
- ✓ **Topic** Quantifiers
- ✓ **Topic** Scope of quantifiers

Topics to be Covered



✓
Topic

Negation of predicate formula

✓
Topic

Sentences with multiple quantifiers

✓
Topic

Relationship diagram

Today's Topic



Topic : Scope of the Quantifier

• The part of the logical expression to which a quantifier can be applied is called the scope of the quantifier.

- A variable whose occurrence is bounded by a quantifier is called a bounded variable. Variables not bounded by any quantifiers are called free variables.

• Generally we represent the scope of the quantifier using brackets or comma otherwise, The Scope of a quantifier is the shortest full sentence/predicate formula which follows it. Everything inside this shortest full sentence is said to be in the scope of the quantifier.



Topic : Scope of the Quantifier

- $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$
- $\exists y(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$
- $(\forall y H(y) \rightarrow \exists z G(z)) \rightarrow \exists z W(z, y)$
- $(\forall z H(y) \rightarrow \exists y W(z, y)) \rightarrow \exists z G(z)$
- $(\forall y H(y) \rightarrow \exists z \exists y W(z, y)) \rightarrow \exists z G(z)$
- $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists x G(z)$



Topic : Scope of the Quantifier

• $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$

Diagram illustrating the scope of quantifiers in the logical expression $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$:

- bounded variable** (blue text): Points to the variable y in $H(y)$ and the variable y in $W(z, y)$.
- bounded Variable** (blue text): Points to the variable z in $W(z, y)$ and the variable z in $G(z)$.
- free Variable** (red text): Points to the variable z in $W(z, y)$ and the variable z in $G(z)$.



Topic : Scope of the Quantifier

$$\exists y (\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists z G(z)$$

$\exists$ will quantify all free variables 'y' in its scope

bounded by $\forall$

bounded

bounded by $\exists y$

bounded

A single variable in a predicate formula can never be quantified using multiple quantifiers

⇓
If variable is under the scope of multiple quantifiers, then it will be quantified using innermost quantifier



H.W.

Topic : Scope of the Quantifier



$$(\forall y H(y) \rightarrow \exists z G(z)) \rightarrow \exists z W(z, y)$$

Diagram illustrating the scope of quantifiers in the logical expression $(\forall y H(y) \rightarrow \exists z G(z)) \rightarrow \exists z W(z, y)$. The expression is annotated with green wavy lines and arrows to show the scope of each quantifier:

- The first $\forall y$ is bounded by a wavy line, and an arrow points from the label "bounded" below to it.
- The $\exists z$ in the second part is bounded by a wavy line, and an arrow points from the label "bounded" below to it.
- The $\exists z$ in the third part is bounded by a wavy line, and an arrow points from the label "bounded" below to it.

free variable



H.W.

Topic : Scope of the Quantifier



$$(\forall z H(y) \rightarrow \exists y W(z, y)) \rightarrow \exists z G(z)$$

Diagram illustrating the scope of quantifiers in the logical expression $(\forall z H(y) \rightarrow \exists y W(z, y)) \rightarrow \exists z G(z)$.

- free**: Points to the variable y in $H(y)$.
- free**: Points to the variable y in $\exists y W(z, y)$.
- bounded**: Points to the variable z in $\forall z H(y)$.
- bounded**: Points to the variable z in $\exists y W(z, y)$.
- bounded**: Points to the variable z in $\exists z G(z)$.



M.W.

Topic : Scope of the Quantifier



$$(\forall y H(y) \rightarrow \exists z \exists y W(z, y)) \rightarrow \exists z G(z)$$

Diagram illustrating the scope of quantifiers in the logical expression $(\forall y H(y) \rightarrow \exists z \exists y W(z, y)) \rightarrow \exists z G(z)$. The expression is annotated with handwritten labels and arrows indicating the scope of each quantifier:

- bounded** (blue) points to the scope of $\forall y$ in $H(y)$.
- bounded** (blue) points to the scope of $\exists z$ in $\exists z \exists y W(z, y)$.
- bounded** (pink) points to the scope of $\exists y$ in $\exists z \exists y W(z, y)$.
- bounded** (orange) points to the scope of $\exists z$ in $\exists z G(z)$.



H.W.

Topic : Scope of the Quantifier



$$(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists x G(z)$$

Diagram illustrating the scope of quantifiers in the logical expression $(\forall y H(y) \rightarrow \exists z W(z, y)) \rightarrow \exists x G(z)$ using handwritten orange annotations:

- bounded**: An arrow points from the label to the variable y in $H(y)$.
- bounded**: An arrow points from the label to the variable z in $W(z, y)$.
- free**: An arrow points from the label to the variable z in $G(z)$.
- free**: An arrow points from the label to the variable x in $\exists x G(z)$.

Q:- Which of the following Predicate formula is/are valid

$$F_1: [\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}] \rightarrow [\forall x \{P(x) \rightarrow S(x)\}]$$

$$F_2: [\forall x \{P(x) \rightarrow S(x)\}] \rightarrow [\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}]$$

(a) F_1 only

(b) F_2 only

(c) Both F_1 & F_2

(d) Neither F_1 nor F_2

Let P : is a politician

S : is a sportsman

$\forall x \{ P(x) \rightarrow S(x) \}$

: for all x ' $P(x) \rightarrow S(x)$ ' is true

* If there exists any ' x ', for which $P(x)$ is true but $S(x)$ is false, then ' $P(x) \rightarrow S(x)$ ' will be false and hence ' $\forall$ ' will return false

* Predicate formula will be true only when $P(x) \rightarrow S(x)$ is true for all x .
Whenever $P(x)$ is false, then $P(x) \rightarrow S(x)$ will always be true. But whenever $P(x)$ is true then $S(x)$ should also be true for that x .

We are checking whether for all ' x ' in the domain $P(x) \rightarrow S(x)$ is true or not
* Truth value of this predicate formula is the truth value returned by ' $\forall$ '

$\forall x \{ P(x) \} \rightarrow \forall x \{ S(x) \}$
LHS R.H.S.

: if $\forall x \{ P(x) \}$ then $\forall x \{ S(x) \}$

i.e., if all are politician then all must be sportsman

* If there is any person in the domain who is not a politician, then $\forall x \{ P(x) \}$ will be false, i.e., L.H.S. of implication will be false $\therefore$ implication will return true

* But if all are politician in the domain, then L.H.S. will be true $\therefore$ for implication to be true, all must be sportsman as well

Truth value of this predicate formula is the truth value returned by implication

$$F_1: [\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}] \rightarrow [\forall x \{P(x) \rightarrow S(x)\}]$$

$U = \{a, b, c\}$

$P(a) = \text{True}$
 $P(b) = \text{False}$
 $P(c) = \text{True}$

at least one false
 $\therefore \forall x \{P(x)\}$ is false

Hence

$\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}$ is true

Hence LHS of implication is true

We know

$P(a) = \text{True}$
 $P(b) = \text{False}$
 $P(c) = \text{True}$

if

$S(a) = \text{False}$
 $S(b) = \text{True}$
 $S(c) = \text{True}$

$\therefore P(a) \rightarrow S(a) = T \rightarrow F = F$
 $P(b) \rightarrow S(b) = F \rightarrow T = T$
 $P(c) \rightarrow S(c) = T \rightarrow T = T$

$\therefore P(x) \rightarrow S(x)$ is false for at least one.
 $\therefore \forall x \{P(x) \rightarrow S(x)\}$ will return false

Hence RHS of implication is false in this case

i.e. it may be possible that
 when LHS is true then RHS becomes false
 $\therefore$ Given Predicate formula is invalid

$$F_2: [\forall x \{ P(x) \rightarrow S(x) \}] \rightarrow [\forall x \{ P(x) \} \rightarrow \forall x \{ S(x) \}]$$

But if $P(x)$ is true for all, then for $P(x) \rightarrow S(x)$ to be true for all, $S(x)$ must also be true for all, then only $\forall x \{ P(x) \rightarrow S(x) \}$ will return true.

if $P(x)$ happens to be false for at least one 'x', then RHS of implication is guaranteed to be true
 ∴ In this case Implication will always return true

i.e. if $\forall x \{ P(x) \rightarrow S(x) \}$ return true
 and $P(x)$ is true for all {i.e. $\forall x \{ P(x) \}$ is true}
 then $S(x)$ is guaranteed to be true for all {i.e. $\forall x \{ S(x) \}$ is true}

Hence
 $\forall x \{ P(x) \} \rightarrow \forall x \{ S(x) \}$
 will also be true

Main implication will always return true
 ∴ Valid Predicate formula

Q:- Which of the following Predicate formula is/are valid

$$F_1: [\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}] \rightarrow [\forall x \{P(x) \rightarrow S(x)\}]$$

$$F_2: [\forall x \{P(x) \rightarrow S(x)\}] \rightarrow [\forall x \{P(x)\} \rightarrow \forall x \{S(x)\}]$$

(a) F_1 only

☒ (b) F_2 only

(c) Both F_1 & F_2

(d) Neither F_1 nor F_2

H.W. Consider following two predicate formula

$$F_1: \quad \forall x \{ P(x) \leftrightarrow Q(x) \} \longrightarrow \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \}$$

$$F_2: \quad \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \} \longrightarrow \forall x \{ P(x) \leftrightarrow Q(x) \}$$

$$U = \{a, b\}$$

$$P(a) = T \text{ \& } Q(a) = F \therefore \forall x \{ Q(x) \} = F$$

$$P(b) = F \text{ \& } Q(b) = T \therefore \forall x \{ P(x) \} = F$$

$\therefore$ L.H.S is true

$$\begin{array}{l} P(a) \leftrightarrow Q(a) \\ T \leftrightarrow F = F \end{array}$$

$\therefore \forall x$ will return false

H.W. Consider following two predicate formula

$$F_1: \quad \forall x \{ P(x) \leftrightarrow Q(x) \} \longrightarrow \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \}$$

$$F_2: \quad \{ \forall x \{ P(x) \} \leftrightarrow \forall x \{ Q(x) \} \} \longrightarrow \forall x \{ P(x) \leftrightarrow Q(x) \}$$

Which of the following is true

- ☒ (a) Only F_1 is valid
- (b) Only F_2 is valid
- (c) Both F_1 and F_2 are valid
- (d) Neither F_1 nor F_2 is valid



Topic : Note



- Let, $P(x)$: x is true.
 $\therefore \sim P(x)$: x is not true / x is false

	Form	Meaning
1	$\forall x P(x)$	All are true / Not at least one is false
2	$\exists x P(x)$	At least one is True / Not all are false
3	$\sim \forall x P(x)$	Not, all are True / At least one is false
4	$\sim \exists x P(x)$	Not, at least one True / All are false
5	$\forall x \sim P(x)$	All are false / Not at least one is True
6	$\exists x \sim P(x)$	At least one false / Not, all are True
7	$\sim \forall x \sim P(x)$	Not, All are false / At least one is True
8	$\sim \exists x \sim P(x)$	Not, At least one false / All are True



Topic : Equivalences



1. $\forall x P(x) \equiv \sim \exists x (\sim P(x))$

$\forall x (\sim (\sim P(x)))$

$\forall x \{P(x)\}$

2. $\exists x P(x) \equiv \sim \forall x (\sim P(x))$

$\exists x \{ \sim (\sim P(x)) \} \equiv \exists x \{ P(x) \}$

3. $\sim \forall x P(x) \equiv \exists x (\sim P(x))$

$\exists x (\sim P(x))$

4. $\sim \exists x P(x) \equiv \forall x (\sim P(x))$

$\forall x \{ \sim P(x) \}$



Topic : Negation of a statement formula

{ When negation sign is propagated inside the
[Predicate formula,

then { Replace $\forall$ by $\exists$

and
replace $\exists$ by $\forall$

and

{ negate the scope of the corresponding quantifier

Q. Negate the predicate formula
 $\forall x \{ P(x) \rightarrow S(x) \}$

Solution

Negation of $\forall x \{ P(x) \rightarrow S(x) \}$

$$\sim \{ \forall x \{ P(x) \rightarrow S(x) \} \}$$

$$\exists x \{ \sim \{ P(x) \rightarrow S(x) \} \}$$

$$\exists x \{ \sim \{ \sim P(x) \vee Q(x) \} \}$$

$$\boxed{\exists x \{ P(x) \wedge \sim Q(x) \}}$$

Q: The predicate formula $\forall x \{ B(x) \wedge I(x) \}$ is equivalent to which of the following predicate formula

- (a) $\exists x \{ B(x) \rightarrow \sim I(x) \}$
- ✓ (b) $\sim \exists x \{ B(x) \rightarrow \sim I(x) \}$
- (c) $\sim \exists x \{ \sim B(x) \rightarrow I(x) \}$
- (d) $\exists x \{ \sim B(x) \rightarrow I(x) \}$

$$A \equiv \sim(\sim A)$$

$$\begin{aligned}\forall x \{ B(x) \wedge I(x) \} &\equiv \sim \{ \sim \forall x \{ B(x) \wedge I(x) \} \} \\ &\equiv \sim \{ \exists x \{ \sim B(x) \vee \sim I(x) \} \} \\ &\equiv \sim \{ \exists x \{ B(x) \rightarrow \sim I(x) \} \} \\ &\equiv \sim \exists x (B(x) \rightarrow \sim I(x)) \\ &\quad \quad \quad \text{|||} \\ &\equiv \sim \exists x \{ I(x) \rightarrow \sim B(x) \}\end{aligned}$$

Sentence with multiple quantifiers $\Rightarrow$

P : "likes", Domain = Set of all students in our class

$P(x, y)$ = x likes $y \equiv y$ is liked by x

① $\forall x \forall y \{P(x, y)\}$: All x likes All y
 $\rightarrow$

② $\forall y \forall x \{P(x, y)\}$: All y liked by All x
 $\rightarrow$

Sentence with multiple quantifiers $\Rightarrow$

P : "likes", Domain = Set of all students in our class

$P(x, y)$ = x likes $y \equiv y$ is liked by x

- ① $\forall x \forall y \{P(x, y)\}$: All x likes All y
Every student in our class likes Every student of class
- ② $\forall y \forall x \{P(x, y)\}$: All y liked by all x
Every student is liked by Every student

$$\forall x \forall y \{P(x, y)\} \equiv \forall y \forall x \{P(x, y)\}$$

Sentence with multiple quantifiers $\Rightarrow$

P : "likes", Domain = Set of all students in our class

$P(x, y)$ = x likes $y \equiv y$ is liked by x

③ $\exists x \forall y \{ P(x, y) \}$: At least one x likes All y
Some one likes Everyone

④ $\forall y \exists x \{ P(x, y) \}$: All y liked by at least one x
Everyone is liked by Some one

$\exists x \forall y \{ P(x, y) \} \Rightarrow \forall y \exists x \{ P(x, y) \}$ is valid
 $\Leftarrow$ not valid

Sentence with multiple quantifiers $\Rightarrow$

P : "likes", Domain = Set of all students in our class

$P(x, y)$ = x likes $y \equiv y$ is liked by x

- ⑤ $\exists y \forall x \{P(x, y)\}$:
 Some one is liked by Everyone
 At least one 'y' is liked by all x
 At least one student in our class is liked by Every student of class
- ⑥ $\forall x \exists y \{P(x, y)\}$:
 Everyone likes somebody
 All x likes some y
 Every student of class likes some one from the class

$\exists y \forall x \{P(x, y)\} \Rightarrow \forall x \exists y \{P(x, y)\}$ is valid
 $\Leftarrow$ need not be true

Sentence with multiple quantifiers $\Rightarrow$

P : "likes", Domain = Set of all students in our class

$P(x, y)$ = x likes $y \equiv y$ is liked by x

⑦ $\exists x \exists y \{P(x, y)\}$: At least one x likes At least one y

⑧ $\exists y \exists x \{P(x, y)\}$: At least one y is liked by at least one x

Sentence with multiple quantifiers $\Rightarrow$

P : "likes", Domain = Set of all students in our class

$P(x, y)$ = x likes $y \equiv y$ is liked by x

⑦ $\exists x \exists y \{P(x, y)\}$: Some one likes someone

⑧ $\exists y \exists x \{P(x, y)\}$: Someone is liked by someone

$$\exists x \exists y \{P(x, y)\} \equiv \exists y \exists x \{P(x, y)\}$$

let Domain is set of all real numbers.

$$\exists x \forall y \{x \cdot y = y\}$$

$$\forall y \exists x (x \cdot y = y)$$

It is true because $x=1$ exist in set of all real no. $\Rightarrow$ then $\Rightarrow$ it is guaranteed to be true

$$\forall y \exists x (x + y = 0)$$

$$\exists x \forall y \{x + y = 0\}$$

for $y=0$, x is '0'
for $y=1$, x is '-1'
for $y=5$, x is '-5'

$\forall y \exists x$ is true

but

$\exists x \forall y$ is not true

$$\textcircled{1} \quad \forall x \forall y \{ P(x, y) \} \equiv \forall y \forall x \{ P(x, y) \}$$

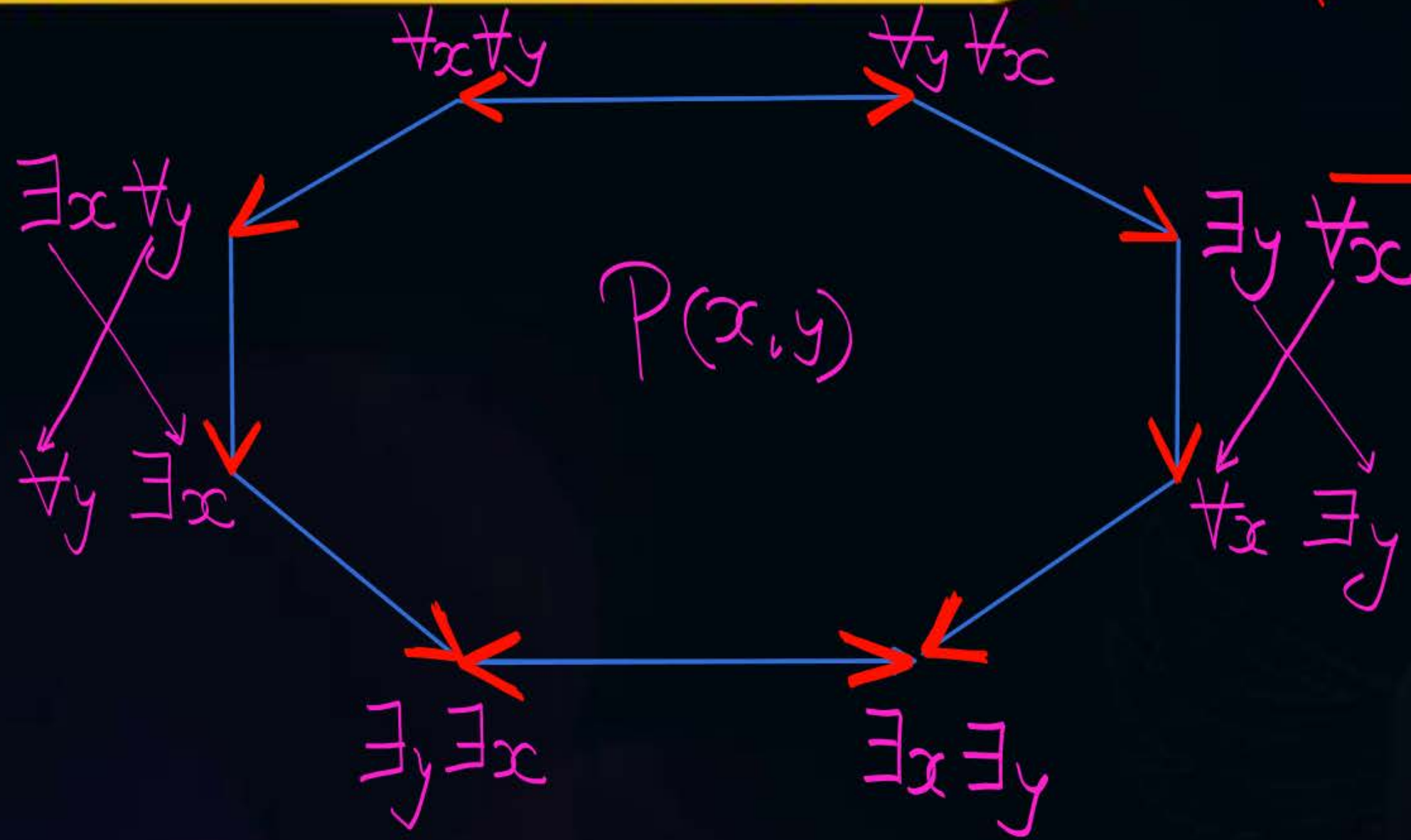
$$\textcircled{2} \quad \exists x \forall y \{ P(x, y) \} \not\equiv \forall y \exists x \{ P(x, y) \}$$

$$\textcircled{3} \quad \exists y \forall x \{ P(x, y) \} \not\equiv \forall x \exists y \{ P(x, y) \}$$

$$\textcircled{4} \quad \exists x \exists y \{ P(x, y) \} \equiv \exists y \exists x \{ P(x, y) \}$$



Topic : Relationship diagram



$\longleftrightarrow$ it will represent Equivalence

$\rightarrow$ it will represent logical implication

#Q. Which of the following is/are true?

A $\forall x \exists y P(x,y) \Rightarrow \exists y \forall x P(x,y)$

B $\forall x \forall y P(x,y) \equiv \exists y \forall x P(x,y)$

☒ **C** $\exists x \forall y P(x,y) \Rightarrow \forall y \exists x P(x,y)$

D $\exists x \exists y P(x,y) \Rightarrow \forall y \exists x P(x,y)$

[MCQ]



#Q. Consider the first order logic sentence $F: \forall x(\exists y R(x, y))$. Assuming non-empty logical domains, which of the sentences below are implied by F ?

✓ I. $\exists y(\exists x R(x, y))$

✗ II. $\exists y(\forall x R(x, y))$

✗ III. $\forall y(\exists x R(x, y))$

✓ IV. $\sim \exists x(\forall y \sim R(x, y))$

$\forall x(\exists y R(x, y)) \equiv F$

A IV only

✓ **B** I and IV only

C II only

D II and III only

Equivalences & Implications

H.W

$$\textcircled{1} \quad \forall x \{ P(x) \wedge Q(x) \} \overset{\textcircled{1}}{\implies} \{ \forall x \{ P(x) \} \wedge \forall x \{ Q(x) \} \} \overset{\textcircled{2}}{\impliedby}$$

$\equiv$

Equivalences & Implications

H.W.

$$\textcircled{2} \quad \exists x \{ P(x) \vee Q(x) \} \overset{\textcircled{1}}{\underset{\textcircled{2}}{\equiv}} \{ \exists x \{ P(x) \} \vee \exists x \{ Q(x) \} \}$$

Equivalences & Implications

HW

$$\textcircled{3} \quad \forall x \{ P(x) \vee Q(x) \} \overset{\textcircled{1}}{\implies} \{ \forall x \{ P(x) \} \vee \forall x \{ Q(x) \} \}$$
$$\overset{\equiv}{\Longleftarrow} \textcircled{2}$$

Equivalences & Implications

HW

$$\textcircled{4} \quad \exists x \{ P(x) \wedge Q(x) \} \overset{\textcircled{1}}{\implies} \{ \exists x \{ P(x) \} \wedge \exists x \{ Q(x) \} \} \overset{\textcircled{2}}{\impliedby}$$

$\equiv$



2 mins Summary



✓
Topic

Negation of predicate formula

✓
Topic

Sentences with multiple quantifiers

✓
Topic

Relationship diagram

THANK - YOU